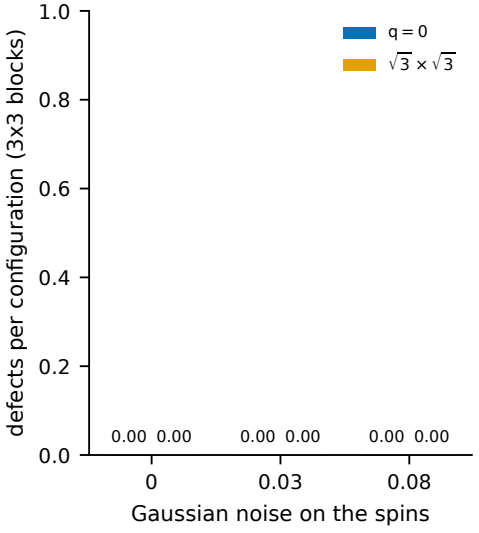
a

classification in the binary dihedral group $2D_3$

none	trivial: frame returns to itself
Z_2	returns up to -1 in the double cover
C_3	triad relabelled cyclically (third-vortex)
C_2	plane normal reversed (half-disclination)

Frame per block: ν = min. eigenvector of Q, triad phase arg $(e^{3i\phi})/3$; candidates ±, 3 cyclic phases; lift along the loop of 4 blocks.

b